

How To Report Evaluations of your LLM: A Story of LLM/Human Misunderstandings

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Abstract

Leaderboard scores for the same model on the same benchmark routinely disagree by tens of points, yet the public record rarely says why. We assemble 46,559 evaluation records from 12 public sources—7,398 models on 1,954 benchmarks—into a single schema and audit what configuration information survives the journey from an evaluation run to a comparison table. In the 28,755-record, 10-source leaderboard subset, the fields that would explain score divergence are absent: prompt template and temperature have zero valid coverage, and none of the leaderboard APIs we examined expose the evaluation harness. Three case studies trace how this missing metadata makes specific cross-source gaps uninterpretable, isolating three distinct mechanisms: a benchmark specification implemented differently by different harnesses (scoring mode, answer parsing); multiple legitimate measurements coexisting under one benchmark name (n-shot, language scope, judge model); and apparent independent agreement that is in fact one pipeline reported twice. A controlled factorial experiment quantifies the underlying effects—prompt format alone drives 27% of GSM8K score variance, while temperature drives under 1%. We propose EvalSpec v0.1, a machine-readable reporting specification whose required fields are grounded in the audit, and release the dataset, schema, and pipeline.

1 Introduction

Open LLM Leaderboard v2 (OLv2) reports Gemma-7B’s score on Big-Bench Hard (BBH) as 21.12. A collection of ArXiv papers, the Jamba, RecurrentGemma, and MAMmoTH2 papers, among others reports BBH scores for the same model between 55 and 60. The gap exceeds 35 percentage points. Neither side is fabricating a result: each reports the score that its pipeline produced. What is missing is the procedure. OLv2 runs BBH with lighteval using log-likelihood scoring over answer

choices, whereas the papers follow the original BBH protocol of 3-shot chain-of-thought generation. Those are not the same measurement, and a benchmark name is not sufficient to make them so. The same pattern recurs for the closely related Gemma-2-9B, whose BBH score we trace in detail in §5.3.1: there, the metadata that would distinguish the two protocols has been stripped from every record before the scores meet on a comparison table.

This example is the rule, not the exception. Evaluation results travel through a chain of hands before a reader encounters them: a model author runs an internal sweep, a leaderboard re-evaluates the checkpoint, an aggregator scrapes the leaderboard, and a downstream paper cites the aggregator. At each stage, information about how the score was produced is lost, dropped, or quietly reinterpreted. By the time two scores meet on a comparison table, the procedures behind them may share nothing more than a benchmark name. For that reason we assemble a dataset of 46,559 evaluation records from 12 sources covering 7,398 models and 1,954 benchmarks, and for a 10-source leaderboard subset of 28,755 records ask what configuration information survives. The fields that most determine how a score is produced—the evaluation harness, the scoring mode, the prompt template, and the n-shot count—are the ones the leaderboard subset records least: prompt template has zero valid coverage, and none of the leaderboard APIs we examined expose the evaluation harness (where harness identity appears it is attached by our pipeline from platform documentation, not returned by the source). Temperature is also absent (zero coverage), but a controlled factorial experiment (§5) shows that temperature drives under 1% of score variance—so its absence costs comparability far less than its zero coverage might suggest, and it is the high-impact fields above whose absence does the damage. We then work through cases where the same model re-

ceived sharply different scores on the same benchmark and diagnose the mechanism that produced each gap.

The case studies are the contribution; our extraction pipeline is the instrument that found them. Each identifies a specific class of mechanism through which benchmark scores become incomparable across sources without either source committing an error: implementation choices left underspecified by the benchmark, multiple legitimate measurements coexisting under one name, and apparent “independent” evidence that shares a single pipeline origin.

Contributions.

- A cross-source audit of metadata coverage in the public evaluation ecosystem.
- Three case studies of cross-source score divergence, diagnosed end-to-end.
- EvalSpec v0.1, a proposed reporting specification grounded in the audit.
- Public release of the full dataset (46,559 records) and the EvalSpec v0.1 vocabulary under CC-BY-4.0, contributed back to the EvalEval shared task (Batzner et al., 2026) as a worked example of the EEE schema in cross-source use; the dataset and EvalSpec are designed to be extended by the community.

2 Related Work

2.1 Evaluation Mispractices and Prior Audits

Several recent studies have shown that benchmark scores are sensitive to implementation choices that the public record rarely exposes. Alzahrani et al. (2024) report score gaps of up to fifteen percentage points on MMLU attributable to harness choice alone, holding the model and benchmark fixed. Biderman et al. (2024) extend this idea to prompt format, tokenisation, and generation configuration, and recommend that these choices be reported as part of any evaluation result. A complementary line of work examines the platforms that host evaluations: Singh et al. (2025) audit Chatbot Arena and document score instability for the same checkpoint under different aliases, along with access asymmetries between providers. Data contamination (Sainz et al., 2023) is a parallel reproducibility threat whose detection likewise depends on richer per-run records.

These studies establish that scores vary with protocol choices. This paper traces the concrete mechanisms through which that variation enters the public record: not which harness choice matters in the abstract, but which specific decisions produce which specific gaps in our cross-source dataset. The missing fields are not exotic: they are quantities known at evaluation time that are simply not exposed downstream.

A separate body of work proposes structured documentation for evaluation artefacts. Model Cards (Mitchell et al., 2019) and Datasheets for Datasets (Gebru et al., 2021) established the norm of structured descriptions for models and data. Three concurrent 2025 proposals extend this to evaluation pipelines: BenchmarkCards (Sokol et al., 2025), EvalCards (Dhar et al., 2025), and Eval Factsheets (Bordes et al., 2025). Each proposes fields that overlap substantially with those identified as absent in our audit. The key difference is that these frameworks are prescriptive: they specify what should be recorded. Our work is empirical first: we establish what the public ecosystem currently records, identify which absences produce uninterpretable divergences, and use that evidence to ground the field selection in EvalSpec v0.1 (Section 6.2).

2.2 The EEE Schema

Our analysis builds on the Every Eval Ever (EEE) schema, introduced by the EvalEval Coalition (Batzner et al., 2026), which articulates the cross-framework score-comparability problem this paper investigates (citing the gap between HELM (Liang et al., 2023) and EleutherAI’s lm-evaluation-harness (Gao et al., 2024) on LLaMA 65B MMLU, 0.637 vs. 0.488); our case studies trace the specific mechanisms behind such gaps in the public record. EEE is a unified JSON format for storing evaluation records from heterogeneous sources. Each record captures one model’s result on one benchmark from one source, organised around four main objects. `model_info` records the model name and unique identifier, with an optional developer field. `eval_library` records the harness name and version. `source_metadata` records the source organisation, source type (documentation or evaluation_run), and the evaluator’s relationship to the model (first_party, third_party, collaborative, or other). `evaluation_results` is an array of result objects, each containing a `metric_config` block (metric name and scoring details) and a

score_details block (score value and uncertainty); a generation_config block (generation arguments such as temperature and prompt template) is optional. A unified schema is the right abstraction for cross-source comparison because the alternative requires implicit alignment that is never written down and never verified. The schema makes alignment explicit and transforms missing information from an invisible gap into a recordable absence: a field that no source populates appears as empty in every record, rather than being silently inferred. It is precisely the fields that are visibly empty across all records that motivate this paper. The dataset is released under CC-BY-4.0.

3 Materials

3.1 How We Gathered the Data

The dataset is assembled from three structurally different classes of source. Leaderboards are taken as authoritative: each leaderboard fetcher downloads its data at runtime and emits a schema-conformant record per model×benchmark cell, with the n-shot count and chain-of-thought flag attached from the leaderboard’s fixed protocol. OLv2 dominates and exposes only model identifiers, scores, and a submission date; its eval_library.name is left as unknown, and no field for evaluation harness, prompt template, scoring mode, or temperature is returned by the API. HuggingFace Model Cards are parsed for the YAML model-index block, capturing first-party results that model authors chose to document. The third source is ArXiv paper extraction: for 100 papers, the HTML rendering is fetched and an LLM (Claude Haiku via OpenRouter) reads paper prose into the EEE schema, returning per-benchmark metadata that may include n-shot count, temperature, prompt template, chain-of-thought flag, and harness name when explicitly stated. The LLM may hallucinate or misattribute values, so we treat these metadata fields as unverified at scale.

3.2 What We Gathered

The full dataset contains 46,559 evaluation records covering 7,398 unique models and 1,954 unique benchmarks across 12 sources. Table 1 reports per-source counts.

The 10 leaderboard sources contribute 28,755 records across 5,301 models and 174 benchmarks. Open LLM Leaderboard v2 (OLv2) dominates, contributing 26,790 records (93.2%) across 4,465

models on 6 benchmarks. HuggingFace Model Cards present the opposite shape: 207 records from only 11 unique models but spanning 156 distinct benchmarks, a handful of model authors documenting many benchmarks each. Chatbot Arena, SWE-Bench, and MT-Bench each report one record per model, reflecting their pairwise or task-specific evaluation structure. The remaining leaderboard sources (AlpacaEval 2, BFCL, BigCodeBench, EvalPlus, WildBench) contribute benchmark-specific coverage within their respective domains.

The ArXiv extraction source contributes a further 17,228 records from 100 papers (counting only the canonical per-benchmark LLM extraction; audit re-extractions are not double-counted). Because each paper typically reports a large baseline table—many comparison models across many benchmarks in addition to the model it introduces—these 100 papers yield 1,726 distinct models and 1,806 distinct benchmarks (Table 1). Papers With Code contributes a further 576 records but is excluded from all statistical analyses due to systematic scraping artefacts (parameter counts parsed as scores). The ArXiv source is excluded from the headline leaderboard-subset metadata coverage analysis (Figure 1) because its metadata fields were LLM-populated rather than read from a structured API; coverage figures would otherwise be inflated by unverified values. The 10-source leaderboard subset measures what platforms expose in machine-readable form, which is the relevant measure of documentation practice. The ArXiv source instead serves the case studies in Section 5, where it provides the cross-source comparison point: a score reported by a leaderboard is compared against the same model×benchmark cell as reported in the original paper.

4 Quantitative Analysis

4.1 Human Annotation of Automatic Score Extractions

Score extraction precision and recall. We audited score extraction against the source papers across 10 ArXiv papers (Gemma 2, Instella, Kimi-Audio, Llama-Embed-Nemotron-8B, Mistral 7B, Phi-3, Phi-4-reasoning-vision, Sailor, SOLAR 10.7B, and TinyLlama), covering 2,027 extracted (model, benchmark, score) entries, and report two complementary metrics. *Precision* — of the scores the pipeline extracted, the fraction

Source	Records	Models	Benchmarks
Open LLM Leaderboard v2	26,790	4,465	LLM-extracted ArXiv records, 16,374 (95.0%)
AlpacaEval 2	446	223	carry a non-null score value. This is a JSON-level
BFCL	327	109	structural check, which confirms that the extraction
BigCodeBench	280	156	pipeline placed a value in the score field and is not
Chatbot Arena	218	218	a verification that the value is semantically correct.
EvalPlus	214	125	
HF Model Card	207	11	
WildBench	122	61	
SWE-Bench	117	117	
MT-Bench	34	34	
<i>Leaderboard subtotal</i>	28,755	5,301	
ArXiv papers (LLM extraction)	17,228	1,726	
Papers With Code	576	383	
Total (12 sources)	46,559	7,398	

Table 1: Per-source record, model, and benchmark counts. The leaderboard subtotal defines the 10-source subset used for metadata coverage analysis; the ArXiv extraction source provides the cross-source comparison point for the case studies. Model and benchmark totals do not sum because many appear in multiple sources.

that match the paper — is **97.5%** (1,977/2,027). The 50 residual errors concentrate in five systematic artefacts: base/chat model-variant duplication, where values reported under a model’s instruction-tuned name are copied onto its base name (Kimi-Audio, 33); a spurious leading “0” read from a “0 X.XX” HumanEval column (TinyLlama, 8); a model-size mislabel of Qwen1.5-32B as “34B” (Gemma 2, 5); duplicate Chatbot-Arena Elo attribution onto base or non-verbatim model names (Gemma 2, 3); and a single wrong pass@k variant (Instella, 1). *Recall* — of the scores the papers report, the fraction the pipeline captured, assessed by reading each paper’s result tables — is approximately **81%** and far more variable (52–100% per paper): the extraction reliably captures a paper’s headline model×benchmark tables but skips secondary ones (per-category breakdowns, safety/responsibility suites, architecture ablations, long-context sub-scores) and, in table-heavy papers, sometimes omits whole comparison models, an effect that worsens with paper length and table count. Per-paper counts are in Table 3 (Appendix A). Together these show that when the pipeline places a value in the score field it usually matches the paper (high precision), but that it does not exhaustively transcribe every reported score (moderate recall); neither check verifies the surrounding metadata (harness, n-shot, metric variant), a harder problem.

Structural presence check. Across the 17,228 LLM-extracted ArXiv records, 16,374 (95.0%) carry a non-null score value. This is a JSON-level structural check, which confirms that the extraction pipeline placed a value in the score field and is not a verification that the value is semantically correct.

4.2 Human Annotation of Automatic Harness Identification

We assembled two balanced samples from the ArXiv extraction source: five papers that name an evaluation harness (*with-harness*: BLOOM, Llama, Sheared LLaMA, Aya 23, Nemotron-4 15B) and five that do not (*without-harness*: Code Llama, Mamba, Reka, Jamba, and a Llama-2-related survey paper), covering 817 with-harness and 869 without-harness harness-call rows. For each paper, the ArXiv HTML was retrieved and the pipeline’s per-row harness assignment was verified against the source text under a strict rubric: the pipeline’s `eval_library.name` value for a row is TRUE only if it correctly names the tool the paper says was used to produce that specific score (with canonical equivalents like “eval-harness” and “Language Model Evaluation Harness” accepted as the same library, but a paper’s distinctly-named fork required as a verbatim match). Methodology is in Appendix B.

On the with-harness sample, per-row harness-ID precision was $712/817 = 87.1\%$. Three of five papers were at 100% precision (BLOOM 436/436, with the *Prompted Language Model Evaluation Harness* fork name correctly extracted; Sheared LLaMA 138/138; Aya 23 58/58). Two papers exposed a real pipeline failure mode: when a paper attributes a named harness to a broad group via a phrase like “across all aforementioned tasks” (Nemotron-4 15B) or “for open access models” (Llemma), the LLM under-extracts and applies the attribution to only the benchmarks explicitly listed in the same sentence, missing the broader group. Nemotron-4 15B precision was $6/96 = 6.2\%$ (only the commonsense-reasoning rows tagged, the remaining math/code/multilingual rows in the same *aforementioned* group left as unknown); Llemma was $74/89 = 83.1\%$.

On the without-harness sample, abstention quality was $869/869 = 100\%$. The pipeline returned unknown for every harness call across all five papers, and exhaustive text search confirmed that none of the five names a public harness for its own evaluations: one (a survey that reports only other

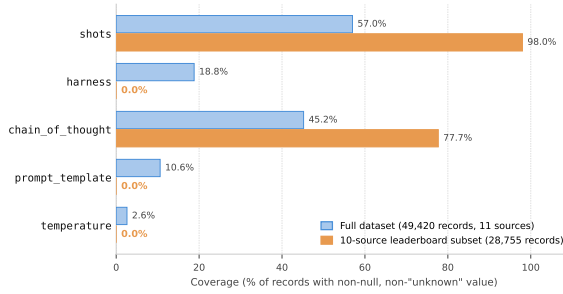


Figure 1: Metadata coverage across five configuration fields, full dataset (46,559 records) vs. the 10-source leaderboard subset (28,755 records). Prompt template and temperature have zero valid coverage in the leaderboard subset. Harness identity shows 93% in the subset, but this is entirely pipeline-attached: OLv2’s API does not expose a harness field, and the pipeline writes `lm_eval` to every OLv2 record from platform documentation. Full numerical values are in Table 6 (Appendix D).

models’ published numbers) runs no evaluations of its own, while the remaining four (Code Llama, Mamba, Jamba, Reka) describe their evaluation setups without naming the tool used.

4.3 Results and Discrepancies

Metadata coverage. Figure 1 reports coverage for five configuration fields across both the full 46,559-record dataset and the 10-source 28,755-record leaderboard subset. Coverage counts a field as populated whenever the EEE record carries a non-“unknown” value, irrespective of whether that value was returned by the source API, attached by the pipeline from platform documentation, or extracted from paper prose by the LLM extractor.

The headline pattern is that the fields most responsible for score divergence are either absent from the leaderboard subset (prompt_template and temperature at 0%) or, in the case of harness identity, populated only because the pipeline attaches `lm_eval` to every Open-LLM-Leaderboard v2 record from platform documentation (§3.1). No leaderboard source we examined returns the harness identity via its API; the 93% leaderboard-subset harness coverage in Figure 1 is the pipeline’s auto-attachment, not source-recorded data. As Section 5.1.1 documents, OLv2 actually runs *lighteval*, and the discrepancy between the pipeline-attached `lm_eval` label and the actual *lighteval* runtime is itself an instance of the same problem: the API exposes nothing, so any attribution downstream is inferred.

Why scoring mode is not in the coverage figure. The coverage audit measures five fields that a source either populates or leaves blank. scoring_mode (log-likelihood vs. free-form generation), the field the case studies show to be most consequential, is not among them because no source in our dataset records it as a discrete field: it is implicit in the harness configuration and recoverable only by knowing the exact tool and task version that produced the score. Its coverage is therefore effectively zero everywhere, for the same reason harness identity is—the value is never exposed—but, unlike harness identity, it cannot even be pipeline-attached from platform documentation, because one platform can run the same benchmark in either mode (§5.1.1). That unmeasurability is itself the argument for promoting scoring_mode to a first-class required field (§6.2).

5 Case Studies

The case studies are organised into three groups corresponding to three distinct mechanisms through which benchmark scores become incomparable across sources; Figure 2 shows one representative divergence per mechanism.

To quantify the relative importance of methodological factors before examining specific failure modes, we ran a controlled factorial experiment crossing four instruction-tuned models (Qwen2.5-14B, Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3) against three benchmarks (BBH, GSM8K, MMLU) under three prompt formats (plain few-shot, instruction-formatted, chain-of-thought), two temperature settings (0.0 and 0.7), and two n-shot counts per benchmark (0-shot plus 3-shot for BBH, 0-shot plus 5-shot for GSM8K and MMLU), with three random seeds per cell; MMLU was restricted to temperature 0.0 and a single seed. The resulting dataset comprises 306 evaluations. Prompt format dominates on GSM8K: at 5-shot, it accounts for approximately 27% of variance ($F = 13.0$, $p < 0.001$; 24 cells per format). MMLU’s reduced 9-cell design at 5-shot is descriptively consistent ($\eta^2 \approx 24.5\%$) but underpowered ($F = 0.98$, $p = 0.43$); the per-format means agree with the GSM8K pattern but cannot independently confirm it. A separate GPQA run (the same four models under the three prompt formats at 5-shot, single seed) shows the same direction — chain-of-thought and instruction formats (both 32.0) over plain (27.5) — with $\eta^2 \approx 13.8\%$, again

underpowered at this scale ($F = 0.72$, $p = 0.51$). Temperature contributes under 1% on GSM8K ($\eta^2 = 0.25\%$); MMLU was run at a single temperature, so the temperature comparison is not available. The n-shot count’s contribution is benchmark-dependent: negligible on MMLU ($\eta^2 \approx 1\%$), but on GSM8K inseparable from the answer parser, because 0-shot runs collapse to zero when the models omit the delimiter the extractor expects (§G.1). Full per-cell results and per-benchmark ANOVAs are in Appendix G. The three case studies below trace this pattern to specific, mechanistically distinct interactions between format choices and evaluation infrastructure.

5.1 Case Study 1: Errors from Undocumented Scoring Mode

The mechanism in this case study: the harness that runs a benchmark makes implementation choices that the benchmark specification leaves underspecified, and different choices produce measurements that cannot be compared on the same scale, even when both are correctly executed.

5.1.1 BBH: Log-Likelihood Scoring vs. Chain-of-Thought Generation

The Big-Bench Hard (BBH) benchmark was introduced by Suzgun et al. (2023) with a specific evaluation protocol: 3-shot chain-of-thought generation, where the model produces free-form reasoning text and the final answer is extracted from that text. This protocol is the standard that ArXiv papers reporting BBH scores follow, and it is the protocol implied by the benchmark’s leaderboard in the literature.

OLv2 evaluates BBH differently. It uses *lighteval* with log-likelihood scoring: for each question, the model is presented with each candidate answer and the answer receiving the highest token log-probability is selected. No text generation occurs, and no chain-of-thought reasoning is elicited. OLv2’s API does not expose the evaluation harness; the harness field in our dataset is empty for all OLv2 records. The actual harness (*lighteval*) and its scoring mode (log-likelihood over answer choices) are confirmed by the platform’s published documentation.

The consequences are visible in the data. Per-source BBH scores for Gemma-7B are in Table 4 (Appendix C), which shows OLv2’s Gemma-7B BBH score of 21.12 against a cluster of ArXiv papers reporting 55.10 (Jamba paper, Recurrent-

Gemma paper), 57.40 (MAMmoTH2 paper), and 59.60.¹ The gap exceeds 35 percentage points. The same pattern holds for Gemma-2-2B (OLv2: 11.76; Instella paper: 40.80) and for MAMmoTH2-7B-Plus (OLv2: 18.93; MAMmoTH2 paper: 63.10).

A controlled scoring-mode comparison anchors this diagnosis. E2 (Appendix G.3) rules out a generic scoring-mode effect: the log-likelihood vs. generation gap on MMLU is at most 0.28 pp across three paired instruction-tuned models, two orders of magnitude smaller than the BBH gap. The same infrastructure drives MMLU-Pro and any other benchmark OLv2 evaluates via *lighteval* log-likelihood; cross-source comparisons involving those scores should be treated with the same caution.

5.1.2 MATH Lvl 5: Answer Parser Replacement

OLv2 evaluates the MATH Level 5 subset (1,324 problems, 5-shot) using a SymPy-based answer parser. In February 2025, HuggingFace replaced this parser with Math-Verify and re-evaluated all submitted models (Kydlicek et al., 2025).

The old parser failed on `\boxed{ }` outputs not matching a specific format string, mishandled sets, matrices, and interval notation via SymPy, and did not implement numerical-equivalence checks (e.g., treating $1/3$ and $0.333\dots$ as different answers). The transition had measurable effects: an average gain of +4.66 percentage points across all 3,751 re-evaluated models; +8.27 points on the Algebra subset; +6.93 points on the Prealgebra subset; and extreme individual cases up to approximately 90 points on those subsets.

Any cross-source comparison that places a pre-transition OLv2 MATH Lvl 5 score alongside a post-transition score, or alongside a score from an external paper that uses a correct parser, is comparing incomparable quantities. OLv2 exposes a model submission date but not an evaluation-execution timestamp, so it is impossible to distinguish pre- and post-transition scores from the record alone: a model submitted in 2024 now carries a post-transition score despite predating the parser change by months.

¹The 59.60 figure is reported under the surface label “BigBench-Hard”. Our normaliser keys on the “BBH” alias, so this value is shown here for completeness but is held out of the automated cross-source collision counts to avoid conflating it with a confirmed alias match; including it would only widen the gap.

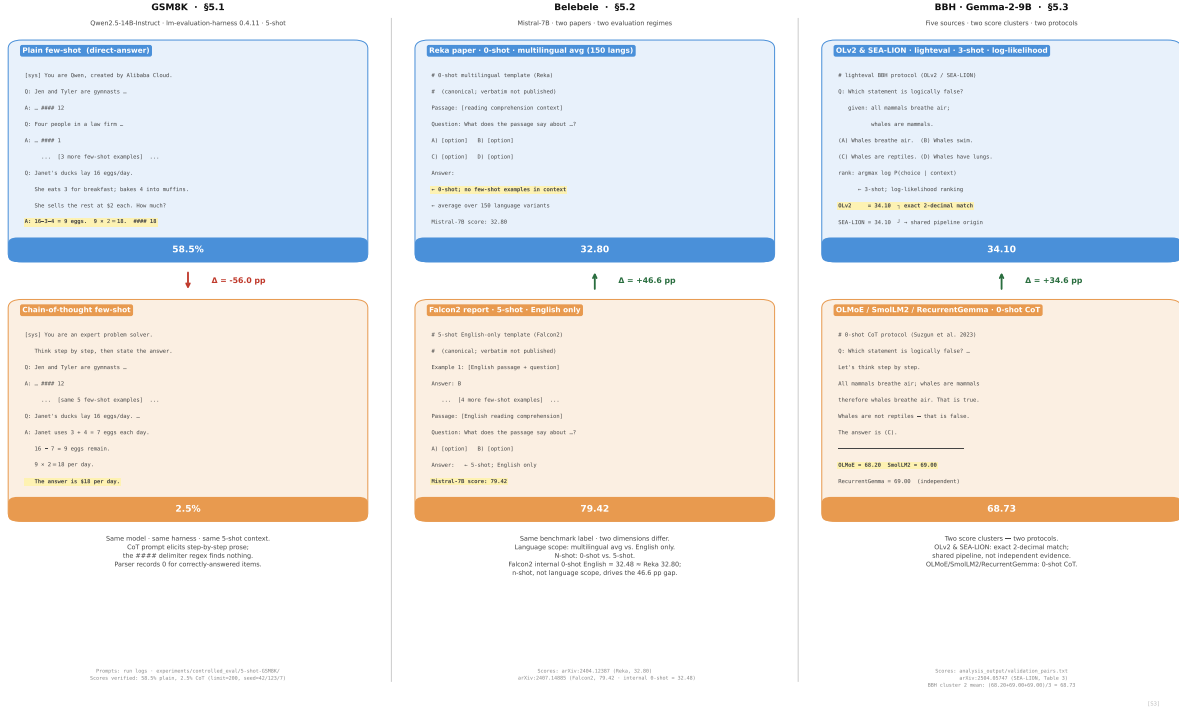


Figure 2: Three mechanisms of score incomparability (§§5.1–5.3). Each column compares two protocol variants under the same benchmark label (blue panel = top configuration, orange = bottom; Δ = bottom – top in percentage points). **Left (§5.1.3)**: GSM8K on Qwen2.5-14B-Instruct, 5-shot, lm-evaluation-harness 0.4.11 — plain vs. chain-of-thought prompt format. CoT elicits prose without the ##### delimiter; parser records 0. Δ = −56.0 pp. **Centre (§5.2.2)**: Belebele on Mistral-7B. Reka paper (Ormazabal et al., 2024): 0-shot, multilingual average (150 languages), 32.80. Falcon2 report (Malartic et al., 2024): 5-shot, English only, 79.42. Falcon2’s internal 0-shot English score (32.48) isolates the n-shot effect. Δ = +46.6 pp. **Right (§5.3.1)**: BBH on Gemma-2-9B across five sources. OLv2 and SEA-LION share a pipeline origin (exact 2-decimal match, 34.10); the Gemma 2/Gemma 3 reports run CoT generation independently (68.20–69.40). Δ = +34.8 pp (cluster gap).

5.1.3 GSM8K: Chain-of-Thought Format and Answer-Parser Failure

The answer-parser sensitivity documented in Section 5.1.2 extends beyond OLv2’s SymPy transition to any harness that applies a format-specific extractor to a model output shaped by the prompt. Within the factorial grid, the Qwen2.5-14B-Instruct $\times$ GSM8K $\times$ 5-shot slice (E1, Appendix G) illustrates this directly: the model is evaluated under three prompt formats using lm-evaluation-harness 0.4.11, with three seeds at temperature 0.0 (Table 2).

Prompt format	Score
plain (direct-answer few-shot)	58.5%
instruct (system-prompt-formatted)	57.0%
cot (chain-of-thought few-shot)	2.5%

Table 2: E1: Qwen2.5-14B-Instruct on GSM8K, 5-shot, temperature 0.0, limit=200. Identical across seeds 42/123/7 (deterministic decoding). Full E1 design in Appendix G.2.

The chain-of-thought format reduces measured accuracy by ≈ 56 percentage points. The mechanism is parser failure (Figure 3): the CoT prompt elicits extended reasoning text, but the harness extracts the final answer by searching for the standard GSM8K delimiter (##### {answer}). When the model produces multi-step reasoning followed by a summary, the regex fails to locate the delimiter and records zero for the item. The model’s reasoning is not incorrect; the parser cannot read the output format the prompt instructed it to produce.

This is the same structural failure as the Math-Verify case: the score reflects the parser’s assumptions about output format, not the model’s mathematical capability. A reader comparing a CoT-prompted score with a plain-prompted score on GSM8K, or a score from one harness whose extractor handles CoT output against one that does not, is comparing incommensurable quantities under an identical benchmark label.

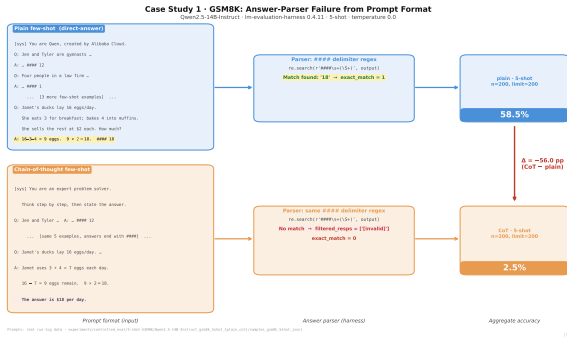


Figure 3: Parser-flow diagram for Case Study 1 (§5.1.3). The same Qwen2.5-14B-Instruct model output is routed through the same #### delimiter regex under two prompt formats. Plain few-shot output contains the delimiter; the regex matches and exact_match= 1. Chain-of-thought output contains step-by-step prose without the delimiter; filtered_resps=‘[invalid]’ and exact_match= 0. Aggregate: 58.5% (plain) vs. 2.5% (CoT), $\Delta = -56.0$ pp.

5.2 Case Study 2: Differences from Implementation Inconsistency

The mechanism in this case study: unlike Case Study 1 (Section 5.1), where a single benchmark specification is implemented differently by different harnesses, here multiple distinct specifications legitimately coexist under the same benchmark name. A reader comparing two scores labeled with the same name may be comparing measurements of genuinely different things.

5.2.1 TruthfulQA: Generation-Based Evaluation at Different N-Shot

TruthfulQA (Lin et al., 2022) includes both multiple-choice metrics (MC1, MC2) and a generation-based metric: the percentage of answers that are both truthful and informative, judged by a GPT-based classifier following the original paper’s protocol. These metrics are not interchangeable, but they operate on different model outputs and different measurement scales.

Our dataset contains records for Llama-2-Chat-7B at 57.04 (one ArXiv paper) and 26.30 (multiple ArXiv papers), all labeled “TruthfulQA”. Verification of the source papers reveals that *both* scores use the generation-based metric (% truthful and informative), not MC1 or MC2:

- The 57.04 figure comes from the Code Llama paper (Rozière et al., 2023) (arXiv:2308.12950), which evaluates Llama-2-Chat-7B as a baseline. The paper

uses a **6-shot** prompt containing six random QA pairs in InstructGPT format, with a GPT-3-based judge. This methodology follows Lin et al. (2022) and is identical in metric to the Llama 2 paper’s evaluation.

- The 26.30 figures come from papers using the Tülu evaluation suite (Iverson et al., 2023), as exemplified by the OLMo paper (Groeneveld et al., 2024) (arXiv:2402.00838). The Tülu suite applies the same generation-based metric at **0-shot**, but with a different judge: a Llama-2-7B fine-tuned classifier rather than the GPT-3-based judge used in the Code Llama evaluation.

The gap (57.04 vs. 26.30) therefore reflects two confounded differences: shot count (6-shot vs. 0-shot) and judge implementation (GPT-3 vs. a Llama-2-7B classifier). Both factors inflate the Code Llama score relative to the Tülu score. Neither source paper records the shot count or the judge model in the metric label; both simply report “TruthfulQA”. Without explicit n_shot and judge fields in the record, the 30-point gap is uninterpretable from the label alone. We isolate the judge factor directly in a controlled experiment (Appendix G.5): holding a fixed set of model responses constant and varying only the judge model shifts mean scores by up to 1.35 points on a 10-point scale and can reverse the ranking of two systems, confirming that judge identity alone— independent of shot count—is enough to make two “TruthfulQA” scores incomparable.

5.2.2 Belebele: N-Shot and Language Scope Confounded

Belebele (Bandarkar et al., 2024) is a multilingual reading-comprehension benchmark. Two evaluation dimensions that should be separately labelled are often not: the language scope (English-only vs. multilingual average) and the n-shot count (which has a substantial effect on accuracy for this benchmark).

Our dataset records Mistral-7B receiving 32.80 from one ArXiv paper and 79.42 from another, both labeled “Belebele”. Verification of the source papers reveals that these two scores differ on both dimensions:

- The 32.80 figure comes from the Reka paper (Ormazabal et al., 2024) (arXiv:2404.12387),

which evaluates Belebele at **0-shot** as a **multilingual average** across 150 language variants. The evaluation was run by the Reka authors themselves (marked with a dagger in the paper).

- The 79.42 figure comes from the Falcon2 technical report (Malartie et al., 2024) (arXiv:2407.14885), which evaluates Belebele at **5-shot** on **English only**.

Crucially, the Falcon2 paper also reports a 0-shot English score for Mistral-7B of 32.48, nearly identical to Reka’s 0-shot multilingual average of 32.80. This internal comparison within the Falcon2 paper isolates the n-shot effect: going from 0-shot to 5-shot on English Belebele yields the full 47-point gain (32.48 → 79.42), while language scope alone (English vs. multilingual average at 0-shot) contributes almost nothing. The dominant driver of the 46.6-point gap between the two labeled scores is therefore the n-shot count, not the language scope.

Neither record carries an n_shot label or a language-scope field. “Belebele 32.80” and “Belebele 79.42” are not comparable measurements of the same thing; they are two different evaluations of Mistral-7B that happen to share a benchmark name.

5.2.3 HellaSwag: Scoring Mode and N-Shot Confounded

HellaSwag (Zellers et al., 2019) is a sentence-completion benchmark for commonsense NLI. The benchmark label conceals two evaluation dimensions at once: scoring mode and n-shot count.

Our dataset records Gemma-7B receiving 49.80 from one ArXiv paper and 81.20 from another, both labeled “HellaSwag”. Verification of the source papers reveals that the scores differ on both dimensions:

- The 49.80 figure comes from the Phi-3 technical report (Abdin et al., 2024) (arXiv:2404.14219), which evaluates Gemma-7B as a comparison baseline at **5-shot** using a **generation-based** pipeline: the model generates free text and the answer is extracted from its output (temperature 0, direct-answer format).
- The 81.20 figure comes from the Gemma technical report (Gemma Team, 2024) (arXiv:2403.08295, Table 6), which reports

Gemma 7B at **0-shot** using **log-likelihood** scoring over the four candidate continuations.

The ≈31-point gap is too large to be explained by the shot-count difference alone: on a log-likelihood benchmark, varying n-shot typically shifts scores by a few points at most, a bound we confirm directly in our factorial grid, where MMLU’s 0-shot and 5-shot log-likelihood means differ by 2.6 pp ($\eta^2 \approx 1\%$; Appendix G.1). The primary driver is the scoring mode. Generation-based evaluation of HellaSwag, asking the model to produce the correct continuation as free text, is structurally harder than log-likelihood evaluation, where the model ranks pre-specified continuations by token probability. Neither paper labels the scoring mode or the n-shot count in the metric label; both simply report “HellaSwag”. Without explicit scoring_mode and n_shot fields in the record, the gap is indistinguishable from a genuine capability difference. Appendix F renders this case as an alternative middle column for Figure 2.

5.3 Case Study 3: Differences from Using a Different Harness

The mechanism in this case study: two sources reporting the same score to multiple decimal places do not necessarily reflect two independent measurements. They can reflect a single measurement that was rereported, making them functionally a single data point disguising as two.

5.3.1 SEA-LION and Gemma-2-9B on BBH

OLv2 reports Gemma-2-9B’s BBH score as 34.10. The SEA-LION paper (Ng et al. 2025; arXiv:2504.05747, IJCNLP-AACL 2025) reports the same model’s BBH score as 34.10, an exact match to two decimal places. The SEA-LION paper lists its benchmark sources as “SEA-HELM Leaderboard, SEACrowd, Open LLM Leaderboard”. An exact two-decimal match of this kind is consistent with either of two mechanisms: direct importation of the OLv2 score for a competitor model, or independent re-evaluation using identical infrastructure (the same lighteval configuration and task suite that OLv2 runs). The EEE record cannot distinguish between them. Our harness-version control (Appendix G.4) shows that independent reruns of the same harness vary by up to 1.3 pp across seeds, so an *exact* two-decimal match is far more consistent with a re-imported score than with an independent re-evaluation, which

would almost never agree to the second decimal. In either case, the two records share a pipeline origin and are not independent measurements of Gemma-2-9B’s BBH capability. The fields that would distinguish replication from re-import—`harness`, `prompt_template_hash`, and `evaluation_timestamp`—were dropped before the records reached the comparison table, leaving the exact two-decimal match as the only visible signal and no way to tell which mechanism produced it.

The Gemma 2 and Gemma 3 technical reports list Gemma-2-9B’s BBH score as 68.20, 69.00, and 69.40 respectively (the Gemma 3 report gives two configurations). These reports use chain-of-thought generation, a different scoring protocol from OLV2’s 3-shot lighteval log-likelihood, explaining the ≈ 35 -point gap. The same harness-mismatch mechanism from Section 5.1.1 applies here as a compounding factor. All five sources and their scores are listed in Table 5 (Appendix C).

6 Discussion

6.1 The Misunderstanding Chain

An evaluation result is produced by a long chain of decisions: which harness, which version, which prompt template, which n-shot count, which temperature, which scoring mode, which answer extractor. At evaluation time, every link in that chain is known. The runtime environment carries the harness version, the prompt template is a string in memory and the temperature is a float in a configuration file. Between evaluation time and the reader, almost all of this information is dropped. Leaderboard APIs return scores and model identifiers but rarely expose the configuration that produced them. Aggregators store whatever the APIs return. Downstream papers cite the aggregator. By the time two scores meet on a comparison table, the procedures behind them are no longer comparable in any verifiable way.

The three case studies in §5 document where this loss occurs: at the leaderboard (§5.1, where OLV2 does not expose harness or scoring mode), at the benchmark label (§5.2, where TruthfulQA, Belebele, and HellaSwag conceal n-shot, metric variant, and scoring mode), and at the appearance of independent measurement (§5.3, where shared pipelines produce apparent replication that is illusory).

Two alternatives to documentation are some-

times proposed. The first is automatic re-extraction from public artefacts. Our extraction pipeline is reliable when the source contains the information: the score-verification audit (§4.1) confirms 97.5% precision across 2,027 row-level annotations on 10 papers (though only $\approx 81\%$ recall — it captures headline tables but skips many secondary ones), and the harness-identification audit (§4.2) achieves 87.1% per-row precision on the with-harness sample (3/5 papers at 100%; the remaining 2 expose a specific failure mode where broad group attributions like “across all aforementioned tasks” are under-extracted) and 100% correct abstention on the without-harness sample. But Figure 1 shows the limit of extraction: in the 10-source leaderboard subset, prompt template and temperature have zero coverage outright, and harness identity is never exposed by a source API—its 93% subset coverage is entirely pipeline-attached from platform documentation (§4.3), not source-recorded. No extraction effort can recover what the source never recorded. The second alternative is independent re-evaluation, which is feasible at small scale but not across the thousands of model-benchmark pairs in a modern aggregated dataset. This leaves documentation: not sufficient in every case, but necessary in the current public ecosystem, because the two alternatives fail precisely where the need is greatest.

6.2 EvalSpec v0.1 as a Proposed Remedy

EvalSpec v0.1 is a machine-readable reporting specification designed to be the smallest field set that turns a score into an identifiable measurement: enough that two records on the same benchmark can be compared without guesswork about the procedure that produced them.

Field selection follows directly from the case studies in §5. `scoring_mode` (log-likelihood vs. free-form generation) is motivated by the BBH label-collapse (§5.1.1) and the HellaSwag analysis (§5.2.3). `harness_name` and `harness_version` are required because the SEA-LION case (§§5.1.1–5.3.1) shows that harness identity is the fingerprint distinguishing genuinely independent measurement from a shared-pipeline artefact. `n_shot` is motivated by the TruthfulQA, Belebele, and HellaSwag audits (§§5.2.1–5.2.3), where the n-shot count shifts the evaluation regime but is absent from every aggregated source’s benchmark label. `benchmark_variant` (recording sub-task keys such as `mc1` vs. `mc2`) is motivated by TruthfulQA and Belebele (§§5.2.1–5.2.2), where the

same name covers metrics with opposite definitions of correctness. `judge_model` is required because TruthfulQA (§5.2.1) uses a fine-tuned judge whose identity is material to the score and absent from the aggregated record; a controlled experiment (Appendix G.5) confirms the effect is not marginal—varying only the judge over fixed responses moves mean scores by up to 1.35 points on a 10-point scale and reorders the system ranking (cross-judge rank correlations span +1.0 to −0.3), so the judge is not an offset that cancels in a comparison. `evaluation_timestamp` is required because the Math-Verify parser replacement (§5.1.2) demonstrates that a silent infrastructure change produces a different score under the same benchmark label. The full field set by tier is in Table 7 (Appendix E).

EvalSpec is encoded as a JSON-LD vocabulary extending `schema.org` and compatible with Croissant metadata records. Compliance is enforced by SHACL shape constraints, so that any record can be mechanically checked against a target tier without human judgement. The prompt template is recorded as a SHA-256 hash of the canonical template string: this keeps the specification zero-overhead for producers while preserving the ability to detect template changes across runs or harness upgrades. A reference validator and the full vocabulary are released alongside the dataset.

None of the Required fields require new instrumentation: the values exist in the runtime environment at evaluation time. The barrier to adoption is not implementation cost but API exposure — leaderboards already record these values internally, but their public interfaces do not return them.

6.3 Recommendations

Leaderboard maintainers. Leaderboard operators should expose `harness_name`, `harness_version`, `scoring_mode`, and `evaluation_timestamp` through their public APIs. These values are already known at evaluation time; adding them is a logging decision, not an engineering project. The SEA-LION case (§5.3.1) shows the cost of not exposing them: an exact score match becomes unreliable evidence of independent replication. Records meeting the EvalSpec Required tier would let downstream researchers separate shared-pipeline artefacts from genuine replications.

Model developers. Model releases should include at least one EvalSpec-R compliant record per reported benchmark. A single well-documented record makes scores verifiable against subsequent third-party evaluations and provides a concrete baseline for detecting benchmark variant drift. E1 (Appendix G.2) shows the magnitude of drift available from a single unrecorded prompt format: 58.5% vs. 2.5% on the same model, benchmark, and harness. The TruthfulQA and Belebele discrepancies (§§5.2.1–5.2.2) show the same conflation at the level of n-shot, metric variant, and judge; a machine-readable record anchors comparison to the configuration actually used in the release.

Schema and platform designers. `scoring_mode` should be treated as a first-class controlled-vocabulary field in any evaluation metadata schema. No combination of other Required fields recovers it: a record can be fully specified on harness, n-shot, and timestamp and still leave the log-likelihood-vs.-generation boundary implicit. The BBH label-collapse (§5.1.1) shows the consequence at ecosystem scale, and E2 (Appendix G.3) shows the converse: when scoring mode is the only varied factor, the gap collapses to under 0.3 pp. A four-value controlled vocabulary (log-likelihood, generation, preference, execution) closes this gap.

7 Conclusions

Public evaluation records routinely report a score without the procedure that produced it. Across 46,559 records from 12 sources, the configuration fields most responsible for score divergence—harness identity, scoring mode, prompt template—are largely absent from the leaderboard subset, where prompt template and temperature reach zero valid coverage and no API exposes the evaluation harness. The cross-source overlap that would allow independent verification is rare and concentrated in a single dominant source. Three case studies show how these absences turn ordinary score gaps into uninterpretable ones, and a controlled factorial experiment confirms that an undocumented prompt-format or answer-parser choice can move a score by tens of points on the same model, benchmark, and harness. Because targeted re-evaluation does not scale and automatic re-extraction cannot recover what a source never recorded, documentation is the remaining identification strategy. EvalSpec v0.1 specifies the minimum set of fields to record,

grounded in the audit and validated against the case studies. The dataset, schema, and pipeline are released for the community to extend.

Limitations

Source concentration. The leaderboard subset is dominated by Open LLM Leaderboard v2, which contributes 26,790 of 28,755 records (93%). Aggregate coverage statistics therefore largely describe OLV2 plus a long tail of much smaller sources; statements about “the leaderboard ecosystem” should be read as statements about one dominant platform and a sparse periphery, not a balanced cross-section. We report the leaderboard subset separately from the full dataset throughout for this reason.

Single-extractor ArXiv pipeline. The ArXiv source is produced by a single LLM extractor (Claude Haiku). Our audits (§4.1, §4.2) verify its output against the source papers on a 20-paper sample by hand, but we did not collect inter-annotator agreement at scale or compare independent extractors, so the reported precision and recall are point estimates on a small sample rather than ecosystem-wide guarantees. We accordingly exclude the ArXiv source from the headline coverage analysis and treat its metadata as unverified at scale.

Case-study selection. The three case studies are hand-selected instances of cross-source divergence, chosen because their mechanisms are diagnosable. They establish that each mechanism occurs and is consequential; they are not a random sample and do not support claims about how frequently each mechanism arises across the ecosystem.

Statistical power. The factorial grid is adequately powered on GSM8K (prompt format $F = 13.0$, $p < 0.001$) but underpowered on MMLU ($F = 0.98$, $p = 0.43$) and GPQA ($F = 0.72$, $p = 0.51$), both reduced to single-seed or single-temperature designs under compute limits. We report the MMLU and GPQA results as directionally consistent with the GSM8K pattern, not as independent confirmation, and rest the load-bearing claims on GSM8K. Several grid cells use a 200-problem subset (`limit=200`) rather than the full benchmark, which supports relative format comparisons but not absolute-score claims.

Pipeline-attached harness labels. The 93% harness coverage in the leaderboard subset is attached

by our pipeline from platform documentation, not returned by any source API. It records what a platform documents it runs, which can itself drift from the runtime (§5.1.1); it is not a source-of-truth field, and we mark it as inferred wherever it appears.

Use of AI Assistants

The evaluation-extraction pipeline central to this paper uses a large language model (Claude Haiku, via OpenRouter) to read ArXiv paper prose into the EEE schema; this methodological use is described in §3.1 and audited in §§4.1–4.2. Separately, the authors used LLM-based assistants for copy-editing and LaTeX formatting during manuscript preparation. All quantitative results, claims, and their interpretation are the authors’ own; AI assistants did not generate experimental data or analysis.

Acknowledgments

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1095	<i>Proceedings of the Conference on Fairness, Account-</i>	2,027 entries across 10 papers, 97.5% precision	1149
1096	<i>ability, and Transparency</i> .	and ≈81% recall overall.	1150

Paper	ArXiv ID	Entries	Precision	Recall
Kimi-Audio	2504.18425	249	86.7%	86.7%
Mistral 7B	2310.06825	62	100%	96.9%
Llama-Embed-Nemotron-8B	2511.07025	280	100%	98.2%
Sailor	2404.03608	205	100%	83.3%
Phi-4-reasoning-vision	2603.03975	150	100%	100%
Instella	2511.10628	322	99.7%	79.7%
SOLAR 10.7B	2312.15166	168	100%	85.7%
TinyLlama	2401.02385	105	92.4%	100%
Phi-3	2404.14219	276	100%	81.1%
Gemma 2	2408.00118	210	96.2%	81.1%
Total		2,027	97.5%	91.1%

Table 3: Per-paper entries, precision (extracted scores that match the paper), and recall (reported scores that were captured) for the score-extraction audit (§4.1).

B Harness-Identification Audit Methodology

For each of the 10 audit papers (§4.2), the ArXiv HTML was retrieved from `arxiv.org/html/` and the pipeline’s per-row `eval_library.name` value was verified against the source text by hand. The audit operates at the (model, benchmark) row level rather than the paper level, because a single paper can use a named harness for some benchmarks and a separate pipeline (machine-translation suite, code sandbox, LLM-as-judge, proof assistant) for others.

Strict-match rubric. A row is TRUE only if the pipeline’s harness label correctly names the tool the paper says was used for that specific (model, benchmark) cell. The rubric accepts canonical equivalents for the same library: “eval-harness”, “lm-evaluation-harness”, “Language Model Evaluation Harness”, and “LM-Evaluation Harness” all refer to EleutherAI’s library and are treated as interchangeable. A paper’s distinctly-named fork (e.g., BLOOM’s *Prompted Language Model Evaluation Harness*, the promptsource-extended fork the BLOOM authors named, built, and released) is required as a verbatim match: the pipeline must produce the fork’s name, not the parent library’s name, because the two tools can produce different scores on the same input.

Sample composition. The *with-harness* pool has five papers (BLOOM, Llemma, Sheared LLaMA, Aya 23, Nemotron-4 15B) covering 817 harness-call rows; the *without-harness* pool has five (Code Llama, Mamba, Reka, Jamba, a Llama-2-related survey paper) covering 869 abstention rows. The two pools are disjoint from the score-verification

(Appendix A). Per-row precision and abstention are reported in §4.2.

C Case-Study Score Tables

Source	Score	Notes
OLv2 (lighteval, log-likelihood)	21.12	3-shot
SEA-LION paper	55.10	CoT generation
Gemma 2 report	55.10	CoT generation
Gemma 3 report	57.40	CoT generation
Gemma 3 report	59.60	CoT generation

Table 4: Gemma-7B BBH scores across sources (§5.1.1). The ≈ 35 -point gap reflects OLv2’s lighteval log-likelihood scoring vs. the ArXiv papers’ chain-of-thought generation. The final row is reported under the surface label “BigBench-Hard”; it is shown for completeness but held out of the automated cross-source collision counts (§5.1.1).

Source	Score	Notes
OLv2 (lighteval, log-likelihood)	34.10	3-shot
SEA-LION paper	34.10	shared pipeline origin
Gemma 2 report	68.20	CoT generation (3-shot)
Gemma 3 report	69.00	CoT generation (0-shot)
Gemma 3 report	69.40	CoT generation

Table 5: Gemma-2-9B BBH scores across five sources (§5.3.1). Two clusters: shared pipeline origin (OLv2/SEA-LION) and protocol mismatch (3-shot log-likelihood vs. 0-shot CoT).

D Metadata Coverage Table

Table 6 reports the numerical values underlying the coverage bar chart in Figure 1 (§4.3).

Field	Full dataset	10-source subset
shots	80.5%	98.0%
harness [†]	63.2%	93.2%
chain_of_thought	50.0%	77.7%
prompt_template	14.1%	0.0%
temperature	3.1%	0.0%

Table 6: Metadata coverage (fraction of records with a non-null, non-“unknown” value). Full dataset: 46,559 records / 12 sources. 10-source subset: 28,755 records, excluding the ArXiv extraction source. Numerical values underlying Figure 1. [†]The harness row counts pipeline-auto-attached values (e.g., `lm_eval` attached to every OLv2 record from platform documentation, not returned by the source API). No leaderboard source exposes harness identity directly.

E EvalSpec v0.1 Field Set

Table 7 lists the complete EvalSpec v0.1 field set by compliance tier, motivated in §6.2.

Tier	Field	Description
Required	model_id	Model identifier (HF path or equivalent)
	benchmark_id	Benchmark name and version
	scoring_mode	{log-likelihood, generation, preference, execution}
	harness_name	Evaluation harness name
	harness_version	Exact harness version string
	n_shot	Number of in-context examples
	benchmark_variant	Sub-task or configuration key
	judge_model	Scorer model (null if not applicable)
	evaluation_timestamp	ISO-8601 date of evaluation run
Recommended	prompt_template_hash	SHA-256 of canonical prompt template
	temperature	Sampling temperature
	top_p	Top- p truncation threshold
	quantisation	Weight quantisation scheme
	hardware	GPU type and count
Full	random_seed	RNG seed
	provenance_url	Link to raw evaluation log or artefact
	contamination_check	Decontamination method and date
	compute_hours	Wall-clock GPU-hours

Table 7: EvalSpec v0.1 field set by compliance tier (§6.2). Required fields are the minimum for score comparability; Recommended fields support reproducibility; Full fields enable complete audit.

F Additional Case-Study Evidence

Figure 4 shows the unified case-study figure with HellaSwag (§5.2.3) as the middle column, an alternative to the Belebele variant used in Figure 2.



Figure 4: Alternative middle column for the unified figure (cf. Figure 2), showing the HellaSwag case study (§5.2.3) instead of Belebele. Phi-3 (5-shot, generation-based) vs. Gemma (0-shot, log-likelihood); both prompts shown are canonical reconstructions since neither source paper publishes its HellaSwag prompts verbatim.

G Experimental Setup

Four experiments and one framing factorial grid anchor the analyses in §5. Table 8 summarises them; per-experiment design and results follow in §§G.1–G.5.

G.1 Factorial Grid

The framing experiment for §5. Four instruction-tuned models (Qwen2.5-72B-Instruct, Qwen2.5-72B-Instruct, Llama-3.1-8B-Instruct, Mistral-7B-Instruct-v0.3) were evaluated on three benchmarks (BBH, GSM8K, MMLU) under three prompt formats (plain few-shot, instruct system-prompt-formatted, cot chain-of-thought few-shot), two temperatures (0.0, 0.7), and two n-shot counts per benchmark: 0-shot plus 3-shot for BBH, 0-shot plus 5-shot for GSM8K and MMLU. Three random seeds (42, 123, 7) were used per cell. MMLU was reduced to temperature 0.0 with a single seed (42) due to evaluation-server constraints. The analysis covers 306 successful evaluations: 144 BBH cells (4 models × 2 n-shots × 3 formats × 2 temperatures × 3 seeds), 144 GSM8K cells (same design), and 18 MMLU cells (3 models × 2 n-shots × 3 formats at temperature 0.0 with a single seed).

Inference was run with lm-evaluation-harness

0.4.11 and vLLM. GSM8K and MMLU runs used a 200-problem subset (limit=200) for compute budget; BBH used the full task suite. Variance attribution uses two separate one-way ANOVAs on the prompt-format factor, computed per benchmark on the 5-shot cells to avoid pooling across the unbalanced GSM8K and MMLU designs. GSM8K contributes 72 cells (4 models × 3 formats × 2 temperatures × 3 seeds): prompt format yields $F = 13.00$, $p = 1.6 \times 10^{-5}$, $\eta^2 \approx 27.4\%$, while temperature contributes $\eta^2 \approx 0.25\%$ ($F = 0.18$, $p = 0.68$). MMLU’s reduced 9-cell design at 5-shot (3 models × 3 formats × 1 temperature × 1 seed) is descriptively consistent ($\eta^2 \approx 24.5\%$) but underpowered ($F = 0.98$, $p = 0.43$); the per-format means agree with the GSM8K pattern but cannot independently confirm it. Per-format means for both benchmarks are reported in Table 9.

We attribute variance to the n-shot factor the same way, pooling each benchmark’s successful cells by shot count (0-shot vs. 5-shot; Table 10), and the two benchmarks behave oppositely in a way that is itself diagnostic. On MMLU the n-shot effect is negligible: pooled across models and formats, 0-shot scores 61.94 and 5-shot 59.34 ($\eta^2 \approx 1.0\%$, $F = 0.16$, $p = 0.69$). This corroboration

ID	Goal	Models	Setup	Result
Factorial Grid	Quantify relative importance of methodological factors	Qwen2.5-14B, Qwen2.5-7B, Llama-3.1-8B, Mistral-7B-v0.3	3 benchmarks $\times$ 3 formats $\times$ 2 temps $\times$ 2 n-shots $\times$ 3 seeds (MMLU reduced to 1 temp $\times$ 1 seed); 312 attempted, 306 completed	Prompt format $\approx 27\%$ of variance on GSM8K ($F=13.0$, $p<0.001$); see §G.1 for the full per-benchmark breakdown
E1	Test parser-failure mechanism under varied prompt format	Qwen2.5-14B-Instruct	GSM8K, 5-shot, lm-evaluation-harness 0.4.11; prompt formats plain / instruct / CoT; temperature 0.0; seeds 42, 123, 7; 200-problem subset	Plain 58.5%, Instruct 57.0%, CoT 2.5% (§5.1.3)
E2	Isolate scoring-mode effect from other factors	Mistral-7B-Instruct-v0.3, Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct (paired); Qwen2.5-14B-Instruct (generation only)	MMLU, 5-shot, plain prompt, temperature 0.0; log-likelihood vs. generation-based scoring	Maximum gap across three paired models: 0.28 pp (§5.1.1, ruling out a generic scoring-mode effect)
E3	Test whether the harness <i>version</i> alone shifts scores	Qwen2.5-1.5B-Instruct, Qwen2.5-7B-Instruct	GSM8K full, 5-shot, hf backend; lm-evaluation-harness 0.4.3 vs. 0.4.11; seeds 42, 123	Cross-version ≤ 0.45 pp, below the seed spread (≤ 1.29 pp); backs harness_version + the SEA-LION argument (§5.3.1)
Judge	Isolate the judge-model effect (LLM-as-judge)	5 contestant models, 4 judges (GPT-4o-mini, Gemini-3.5-Flash, Llama-3.1-70B, Claude-Haiku)	MT-Bench turn-1; identical responses scored by each judge	Judge-induced spread up to 1.35 pts/10; ranking flips (cross-judge Spearman +1.0 to -0.3); see §G.5

Table 8: All experiments anchoring the analyses in §5. The Factorial Grid is the framing experiment; E1, E2, E3, and the judge-model experiment each isolate a specific mechanism. Per-cell results and full configurations are in the subsections below.

rates the HellaSwag analysis (§5.2.3) with a controlled measurement: on a log-likelihood-scored benchmark, varying n-shot moves the score by at most a few points. On GSM8K the n-shot factor appears dominant ($\eta^2 \approx 65\%$ pooled, 80% within plain), but the effect is not a capability gain: every 0-shot cell scores exactly 0.0, because the instruction-tuned models without few-shot priming do not emit the ##### answer delimiter the GSM8K extractor requires, so the parser records zero for every item. This is the same parser-format coupling documented in Case Study 1 (§5.1.3), here triggered by shot count rather than prompt format: on a generation-and-parse benchmark n-shot and scoring interact, whereas on a log-likelihood benchmark n-shot is nearly inert.

G.2 E1: GSM8K Prompt-Format Sensitivity

A slice of the Factorial Grid isolating the Qwen2.5-14B-Instruct $\times$ GSM8K $\times$ 5-shot cells at temperature 0.0. Three prompt formats are compared with seeds 42, 123, 7 (nine runs total; Table 11):

The same nine cells run at temperature 0.7 give means across three seeds of 58.67% (plain), 52.67% (instruct), and 2.83% (CoT), confirming

Benchmark	Format	n	Mean	Range
GSM8K	plain	24	38.75	18.50–59.50
GSM8K	instruct	24	34.19	16.50–57.00
GSM8K	cot	24	19.25	2.00–41.00
MMLU	plain	3	63.60	55.17–73.18
MMLU	instruct	3	62.45	52.52–72.46
MMLU	cot	3	51.98	35.97–61.95

Table 9: Per-benchmark, per-format score distributions at 5-shot in the factorial grid. GSM8K cells are pooled across 4 models $\times$ 2 temperatures $\times$ 3 seeds; MMLU cells across 3 models $\times$ 1 temperature $\times$ 1 seed. The plain/instruct/cot ordering of means is the same on both benchmarks; the GSM8K plain-cot gap is 19.50 pp and the MMLU plain-cot gap is 11.61 pp. Raw values released with the dataset.

that the parser-failure mechanism holds across sampling regimes.

G.3 E2: MMLU Scoring-Mode Control

Three instruction-tuned models were evaluated on MMLU 5-shot under both log-likelihood and generation-based scoring to test whether scoring mode alone produces large score gaps (Table 12). A fourth model (Qwen2.5-14B-Instruct) contributes

Benchmark	n-shot	n	Mean
GSM8K	0-shot	72	0.00
GSM8K	5-shot	72	30.73
MMLU	0-shot	9	61.94
MMLU	5-shot	9	59.34

Table 10: Per-benchmark, per-shot score means, pooled across models, prompt formats, temperatures, and seeds. One-way ANOVA on the n-shot factor: GSM8K $\eta^2 \approx 65.1\%$ ($F = 264.6$, $p < 10^{-30}$), MMLU $\eta^2 \approx 1.0\%$ ($F = 0.16$, $p = 0.69$). The GSM8K 0-shot mean of exactly 0.00 reflects parser failure, not absent capability: the instruction-tuned models omit the ##### answer delimiter without few-shot priming (§5.1.3), so the apparent n-shot effect is the same parser-format coupling as Case Study 1. On MMLU (log-likelihood) the n-shot effect is negligible.

Prompt format (temp 0.0, seeds 42/123/7)	Score
plain (direct-answer few-shot)	58.5%
instruct (system-prompt-formatted)	57.0%
cot (chain-of-thought few-shot)	2.5%

Table 11: Qwen2.5-14B-Instruct on GSM8K, 5-shot, lm-evaluation-harness 0.4.11, limit=200. Scores at temperature 0.0 are identical across all three seeds (deterministic decoding), so each row reports a single number rather than a mean with variance. The cot collapse to 2.5% reflects parser failure: the standard GSM8K extractor `re.search(r'#####s+(\S+)', output)` finds no delimiter in the chain-of-thought free-text output (§5.1.3).

a generation-only run for reference.

G.4 E3: Harness-Version Sensitivity

To isolate the effect of the evaluation-harness *version*, two instruction-tuned models (Qwen2.5-1.5B-Instruct, Qwen2.5-7B-Instruct) were evaluated on the full GSM8K test set (5-shot, hf backend) under two versions of lm-evaluation-harness, 0.4.3 and 0.4.11, with two random seeds (42, 123). The model, benchmark, and all generation settings are held fixed, so the only variable is the harness version (Table 13).

The harness version shifts the GSM8K score by less than the run-to-run seed variation: cross-version differences are at most 0.45 pp, while seeds alone move the same configuration by up to 1.29 pp. The effect is real but small relative to seed noise, which is precisely why `harness_version` belongs in the EvalSpec Required tier (§6.2)—so that a future divergence can be attributed to the version rather than guessed at. The same control

Model	log-likelihood	generation	$ \Delta $ (pp)
Mistral-7B-Instruct-v0.3	61.92	61.64	0.28
Qwen2.5-7B-Instruct	74.26	74.43	0.17
Llama-3.1-8B-Instruct	68.30	68.16	0.14
Qwen2.5-14B-Instruct	—	79.79	—

Table 12: MMLU 5-shot, plain prompt, temperature 0.0, lm-evaluation-harness 0.4.11. Log-likelihood runs use a 500-question subset (limit=500); generation runs use the full MMLU test set (limit=14042). The maximum gap across the three paired models is 0.28 pp, which rules out a generic scoring-mode effect: the ≈ 35 -point BBH gap documented in §5.1.1 is specific to the interaction between lighteval’s log-likelihood approach and BBH’s chain-of-thought structure, not a property of log-likelihood scoring in general.

Model	Harness	seed 42	seed 123
Qwen2.5-1.5B-Instruct	0.4.3	32.83	33.74
Qwen2.5-1.5B-Instruct	0.4.11	33.06	33.36
Qwen2.5-7B-Instruct	0.4.3	75.06	76.35
Qwen2.5-7B-Instruct	0.4.11	75.51	—

Table 13: GSM8K (full test set, 5-shot, hf backend) under lm-evaluation-harness 0.4.3 vs. 0.4.11 with identical configuration. The largest seed-matched cross-version difference is 0.45 pp (Qwen2.5-7B, seed 42), below the within-version seed spread (up to 1.29 pp on Qwen2.5-7B/0.4.3). The 0.4.11 / 7B seed-123 cell is omitted. Raw records released with the dataset.

bounds the run-to-run variability invoked in the SEA-LION analysis (§5.3.1).

G.5 Judge-Model Sensitivity

Motivated by the TruthfulQA judge confound (§5.2.1) and the four LLM-judge sources in the dataset (Chatbot Arena, AlpacaEval 2, MT-Bench, WildBench), this experiment isolates the effect of the judge model. Five instruction-tuned “contestant” models each generated one response to the 80 turn-1 MT-Bench questions (Zheng et al., 2023); the identical responses were then scored by four judge models from different providers (GPT-4o-mini, Gemini-3.5-Flash, Llama-3.1-70B-Instruct, Claude-Haiku-4.5) using the canonical MT-Bench single-answer 1–10 grading rubric at temperature 0. Because the responses are held fixed, any variation in a contestant’s mean score across judges is attributable to the judge alone. Each cell aggregates 78–80 ratings, except Gemini-3.5-Flash (48–63), which intermittently returned empty completions.

Table 14 reports the mean rating per (contestant, judge). The judge-induced spread—the gap between a contestant’s highest and lowest judge

mean—averages 1.08 points and reaches 1.35 points on the 10-point scale. More consequentially, the judge choice reorders the leaderboard: Ministral-8B is ranked first by GPT-4o-mini and Llama-3.1-70B but last by Gemini-3.5-Flash. Pairwise Spearman rank correlations between judges span the full usable range, from +1.00 (GPT-4o-mini vs. Llama-3.1-70B, identical order) down to −0.30 (GPT-4o-mini vs. Gemini-3.5-Flash, anti-correlated). The judge model is therefore not a calibration offset that cancels in a comparison; it can invert which system appears better, so a benchmark score produced by an LLM judge is uninterpretable without recording the judge (§6.2).

Contestant	GPT-4o-m	Gem-3.5	Llama-70B	Cl-Haiku	spread
Gemma-3-4B	8.24	7.96	8.31	7.26	1.04
Qwen2.5-7B	7.86	8.02	7.96	6.97	1.05
Llama-3.1-8B	8.04	7.61	8.22	6.87	1.35
Ministral-8B	8.44	7.54	8.62	7.51	1.11
Qwen2.5-72B	8.32	8.59	8.38	7.74	0.84

Table 14: Mean MT-Bench turn-1 rating (1–10) per contestant under each judge; identical responses are scored by all four judges. “spread” is the max–min judge mean for that contestant (mean 1.08, max 1.35). Ministral-8B ranks first under GPT-4o-mini and Llama-70B but last under Gemini; cross-judge Spearman rank correlation ranges from +1.00 to −0.30. Raw ratings released with the dataset.